

Proof: Consider the mapping cylinder M_i , the quotient of $A \times I \amalg B$ in which $(a, 1)$ is identified with $i(a)$. Let $g_t : A \rightarrow M_i$ be the homotopy mapping $a \in A$ to the image of $(a, 1 - t) \in A \times I$ in M_i , and let $\tilde{g}_0$ be the inclusion $B \hookrightarrow M_i$. The cofibration property gives $\tilde{g}_t : B \rightarrow M_i$ with $\tilde{g}_t i = g_t$. Restricting to a fixed $t > 0$, this implies i is injective since g_t is. Furthermore, since g_t is a homeomorphism onto its image $A \times \{1 - t\}$, the relation $\tilde{g}_t i = g_t$ implies that the map $g_t^{-1} \tilde{g}_t : i(A) \rightarrow A$ is a continuous inverse of $i : A \rightarrow i(A)$. $\square$

Many constructions for fibrations have analogs for cofibrations, and vice versa. For example, for an arbitrary map $f : A \rightarrow B$ the inclusion $A \hookrightarrow M_f$ is readily seen to be a cofibration, so the analog of the factorization $A \hookrightarrow E_f \rightarrow B$ of f into a homotopy equivalence followed by a fibration is the factorization $A \hookrightarrow M_f \rightarrow B$ into a cofibration followed by a homotopy equivalence. Even the definition of M_f is in some way dual to the definition of E_f , since E_f can be defined as a pullback and M_f can be defined as a dual pushout. In general, the **pushout** of maps $f : Z \rightarrow X$ and $g : Z \rightarrow Y$ is defined as the quotient of $X \amalg Y$ under the identifications $f(z) \sim g(z)$.

$$\begin{array}{ccc} E_f & \longrightarrow & B^I \\ \downarrow & & \downarrow \\ A & \longrightarrow & B \end{array} \quad \begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ A \times I & \longrightarrow & M_f \end{array}$$

Thus the pushout is a quotient of $X \amalg Y$, while the pullback of maps $X \rightarrow Z$ and $Y \rightarrow Z$ is a subobject of $X \times Y$, so we see here two instances of duality: a duality between disjoint union and product, and a duality between subobjects and quotients. The first of these is easily explained, since a collection of maps $X_\alpha \rightarrow X$ is equivalent to a map $\coprod_\alpha X_\alpha \rightarrow X$, while a collection of maps $X \rightarrow X_\alpha$ is equivalent to a map $X \rightarrow \prod_\alpha X_\alpha$. The notation $\coprod$ for the ‘coproduct’ was chosen to indicate that it is dual to $\prod$. If we were dealing with basepointed spaces and maps, the coproduct would be wedge sum. In the category of abelian groups the coproduct is direct sum.

The duality between subobjects and quotient objects is clear for abelian groups, where subobjects are kernels and quotient objects cokernels. The strict topological analog of a kernel is a fiber of a fibration. Dually, the topological analog of a cokernel is the **cofiber** B/A of a cofibration $A \hookrightarrow B$. If we make an arbitrary map $f : A \rightarrow B$ into a cofibration $A \hookrightarrow M_f$, the cofiber is the mapping cone $C_f = M_f / (A \times \{0\})$.

In the diagram showing E_f and M_f as pullback and pushout, there also appears to be some sort of duality involving the terms $A \times I$ and B^I . This leads us to ask whether $X \times I$ and X^I are in some way dual. Indeed, if we ignore topology and just think set-theoretically, this is an instance of the familiar product-coproduct duality since the product of copies of X indexed by I is X^I , all functions $I \rightarrow X$, while the coproduct of copies of X indexed by I is $X \times I$, the disjoint union of the sets $X \times \{t\}$ for $t \in I$. Switching back from the category of sets to the topological category, we can view X^I as a ‘continuous product’ of copies of X and $X \times I$ as a ‘continuous coproduct.’

On a less abstract level, the fact that maps $A \times I \rightarrow B$ are the same as maps $A \rightarrow B^I$ indicates a certain duality between $A \times I$ and B^I . This leads to a duality between

suspension and loop space, since ΣA is a quotient of $A \times I$ and ΩB is a subspace of B^I . This duality is expressed in the adjoint relation $\langle \Sigma X, Y \rangle = \langle X, \Omega Y \rangle$ from §4.3. Combining this duality between Σ and Ω with the duality between fibers and cofibers, we see a duality relationship between the fibration and cofibration sequences of §4.3:

$$\begin{aligned} \cdots \rightarrow \Omega F \rightarrow \Omega E \rightarrow \Omega B \rightarrow F \rightarrow E \rightarrow B \\ A \rightarrow X \rightarrow X/A \rightarrow \Sigma A \rightarrow \Sigma X \rightarrow \Sigma(X/A) \rightarrow \cdots \end{aligned}$$

Pushout and pullback constructions can be generalized to arbitrary diagrams. In the case of pushouts, this was done in §4.G where we associated a space $\sqcup X$ to a diagram of spaces X . This was the quotient of the coproduct $\coprod_v X_v$, with v ranging over vertices of the diagram, under the identifications $x \sim f_e(x)$ for all maps f_e associated to edges e of the diagram. The dual construction $\prod X$ would be the subspace of the product $\prod_v X_v$ consisting of tuples (x_v) with $f_e(x_v) = x_w$ for all maps $f_e: X_v \rightarrow X_w$ in the diagram. Perhaps more useful in algebraic topology is the homotopy variant of this notion obtained by dualizing the definition of ΔX in the previous section. This is the space ∇X consisting of all choices of a point x_v in each X_v and a path y_e in the target space of each edge map $f_e: X_v \rightarrow X_w$, with $y_e(0) = f(x_v)$ and $y_e(1) = x_w$. The subspace with all paths constant is $\prod X$. In the case of a diagram $\cdots \rightarrow X_2 \rightarrow X_1$ such as a Postnikov tower this construction gives something slightly different from simply turning each successive map into a fibration via the usual pathspace construction, starting with $X_2 \rightarrow X_1$ and proceeding up the tower, as we did in §4.3. The latter construction is rather the dual of an iterated mapping cylinder, involving spaces of maps $\Delta^n \rightarrow X_v$ instead of simply pathspaces. One could use such mapping spaces to generalize the definition of ∇X from diagrams of spaces to complexes of spaces.

As special cases of the constructions $\sqcup X$ and $\prod X$ we have direct limits and inverse limits for diagrams $X_0 \rightarrow X_1 \rightarrow \cdots$ and $\cdots \rightarrow X_1 \rightarrow X_0$, respectively. Since inverse limit is related to product and direct limit to coproduct, it is common practice in some circles to use reverse logic and call inverse limit simply ‘limit’ and direct limit ‘colimit.’ The homotopy versions are then called ‘holim’ for ∇X and ‘hocolim’ for ΔX . This terminology is frequently used for more general diagrams as well.

Homotopy Groups with Coefficients

There is a somewhat deeper duality between homotopy groups and cohomology, which one can see in the fact that cohomology groups are homotopy classes of maps into a space with a single nonzero homotopy group, while homotopy groups are homotopy classes of maps from a space with a single nonzero cohomology group. This duality is in one respect incomplete, however, in that the cohomology statement holds for an arbitrary coefficient group, but we have not yet defined homotopy groups with coefficients. In view of the duality, one would be tempted to define $\pi_n(X; G)$ to be the set of basepoint-preserving homotopy classes of maps from the cohomology analog of a Moore space $M(G, n)$ to X . The cohomology analog of $M(G, n)$ would be a space

Y whose only nonzero cohomology group $\tilde{H}^i(Y; \mathbb{Z})$ is G for $i = n$. Unfortunately, such a space does not exist for arbitrary G , for example for $G = \mathbb{Q}$, since we showed in Proposition 3F.12 that if the cohomology groups of a space are all countable, then they are all finitely generated.

As a first approximation to $\pi_n(X; G)$ let us consider $\langle M(G, n), X \rangle$, the set of basepoint-preserving homotopy classes of maps $M(G, n) \rightarrow X$. To give this set a more suggestive name, let us call it $\mu_n(X; G)$. We should assume $n > 1$ to guarantee that the homotopy type of $M(G, n)$ is well-defined, as shown in Example 4.34. For $n > 1$, $\mu_n(X; G)$ is a group since we can choose $M(G, n)$ to be the suspension of an $M(G, n-1)$. And if $n > 2$ then $\mu_n(X; G)$ is abelian since we can choose $M(G, n)$ to be a double suspension.

There is something like a universal coefficient theorem for these groups $\mu_n(X; G)$:

Proposition 4H.2. *For $n > 1$ there are natural short exact sequences*

$$0 \rightarrow \text{Ext}(G, \pi_{n+1}(X)) \rightarrow \mu_n(X; G) \rightarrow \text{Hom}(G, \pi_n(X)) \rightarrow 0.$$

The similarity with the universal coefficient theorem for cohomology is apparent, but with a reversal of the variables in the Ext and Hom terms, reflecting the fact that $\mu_n(X; G)$ is covariant as a functor of X and contravariant as a functor of G , just like the Ext and Hom terms.

Proof: Let $0 \rightarrow R \xrightarrow{i} F \rightarrow G \rightarrow 0$ be a free resolution of G . The inclusion map i is realized by a map $M(R, n) \rightarrow M(F, n)$, where $M(R, n)$ and $M(F, n)$ are wedges of S^n 's corresponding to bases for F and R . Converting this map into a cofibration via the mapping cylinder, the cofiber is an $M(G, n)$, as one sees from the long exact sequence of homology groups. As in §4.3, the cofibration sequence

$$M(R, n) \rightarrow M(F, n) \rightarrow M(G, n) \rightarrow M(R, n+1) \rightarrow M(F, n+1)$$

gives rise to the exact sequence across the top of the following diagram:

$$\begin{array}{ccccccc} \mu_{n+1}(X; F) & \longrightarrow & \mu_{n+1}(X; R) & \longrightarrow & \mu_n(X; G) & \longrightarrow & \mu_n(X; F) \longrightarrow \mu_n(X; R) \\ \parallel & & \parallel & & & & \parallel \\ \text{Hom}(F, \pi_{n+1}(X)) & \xrightarrow{i^*} & \text{Hom}(R, \pi_{n+1}(X)) & & \text{Hom}(F, \pi_n(X)) & \xrightarrow{i^*} & \text{Hom}(R, \pi_n(X)) \end{array}$$

The four outer terms of the exact sequence can be identified with the indicated Hom terms since mapping a wedge sum of S^n 's into X amounts to choosing an element of $\pi_n(X)$ for each wedge summand. The kernel and cokernel of i^* are $\text{Hom}(G, -)$ and $\text{Ext}(G, -)$ by definition, and so we obtain the short exact sequence we are looking for. Naturality will be left for the reader to verify. $\square$

Unlike in the universal coefficient theorems for homology and cohomology, the short exact sequence in this proposition does not split in general. For an example, take $G = \mathbb{Z}_2$ and $X = M(\mathbb{Z}_2, n)$ for $n \geq 2$, where the identity map of $M(\mathbb{Z}_2, n)$

defines an element of $\mu_n(M(\mathbb{Z}_2, n); \mathbb{Z}_2) = \langle M(\mathbb{Z}_2, n), M(\mathbb{Z}_2, n) \rangle$ having order 4, as we show in Example 4L.7, whereas the two outer terms in the short exact sequence can only contain elements of order 2 since $G = \mathbb{Z}_2$. This example shows also that $\mu_n(X; \mathbb{Z}_m)$ need not be a module over $\mathbb{Z}_m$, as homology and cohomology groups with $\mathbb{Z}_m$ coefficients are.

The proposition implies that the first nonzero $\mu_i(S^n; \mathbb{Z}_m)$ is $\mu_{n-1}(S^n; \mathbb{Z}_m) = \mathbb{Z}_m$, from the Ext term. This result would look more reasonable if we changed notation to replace the subscript $n - 1$ by n . So let us make the definition

$$\pi_n(X; \mathbb{Z}_m) = \langle M(\mathbb{Z}_m, n - 1), X \rangle = \mu_{n-1}(X; \mathbb{Z}_m)$$

There are two good reasons to expect this to be the right definition. The first is formal: $M(\mathbb{Z}_m, n - 1)$ is a ‘cohomology $M(\mathbb{Z}_m, n)$ ’ since its only nontrivial cohomology group $\tilde{H}^i(M(\mathbb{Z}_m, n - 1); \mathbb{Z})$ is $\mathbb{Z}_m$ in dimension n . The second reason is more geometric: Elements of $\pi_n(X; \mathbb{Z}_m)$ should be homotopy classes of ‘homotopy-theoretic cycles mod m ,’ meaning maps $D^m \rightarrow X$ whose boundary is not necessarily a constant map as would be the case for $\pi_n(X)$, but rather whose boundary is m times a cycle $S^{n-1} \rightarrow X$. This is precisely what a map $M(\mathbb{Z}_m, n - 1) \rightarrow X$ is, if we choose $M(\mathbb{Z}_m, n - 1)$ to be S^{n-1} with a cell e^n attached by a map of degree m .

Besides the calculation $\pi_n(S^n; \mathbb{Z}_m) \approx \mathbb{Z}_m$, the proposition also yields an isomorphism $\pi_n(M(\mathbb{Z}_m, n); \mathbb{Z}_m) \approx \text{Ext}(\mathbb{Z}_m, \mathbb{Z}_m) = \mathbb{Z}_m$. Both these results are in fact special cases of a Hurewicz-type theorem relating $\pi_n(X; \mathbb{Z}_m)$ and $H_n(X; \mathbb{Z}_m)$, which is proved in [Neisendorfer 1980].

Along with $\mathbb{Z}$ and $\mathbb{Z}_m$, another extremely useful coefficient group for homology and cohomology is $\mathbb{Q}$. We pointed out above the difficulty that there is no cohomology analog of $M(\mathbb{Q}, n)$. The groups $\mu_n(X; \mathbb{Q})$ are also problematic. For example the proposition gives $\mu_{n-1}(S^n; \mathbb{Q}) \approx \text{Ext}(\mathbb{Q}, \mathbb{Z})$, which is a somewhat complicated uncountable group as we showed in §3.F. However, there is an alternative approach that turns out to work rather well. One defines rational homotopy groups simply as $\pi_n(X) \otimes \mathbb{Q}$, analogous to the isomorphism $H_n(X; \mathbb{Q}) \approx H_n(X; \mathbb{Z}) \otimes \mathbb{Q}$ from the universal coefficient theorem for homology. See [SSAT] for more on this.

Homology Decompositions

Eckmann–Hilton duality can be extremely helpful as an organizational principle, reducing significantly what one has to remember, and providing valuable hints on how to proceed in various situations. To illustrate, let us consider what would happen if we dualized the notion of a Postnikov tower of principal fibrations, where a space is represented as an inverse limit of a sequence of fibers of maps to Eilenberg–MacLane spaces. In the dual representation, a space would be realized as a direct limit of a sequence of cofibers of maps from Moore spaces.

In more detail, suppose we are given a sequence of abelian groups G_n , $n \geq 1$, and we build a CW complex X with $H_n(X) \approx G_n$ for all n by constructing inductively

an increasing sequence of subcomplexes $X_1 \subset X_2 \subset \cdots$ with $H_i(X_n) \approx G_i$ for $i \leq n$ and $H_i(X_n) = 0$ for $i > n$, where:

- (1) X_1 is a Moore space $M(G_1, 1)$.
- (2) X_{n+1} is the mapping cone of a cellular map $h_n: M(G_{n+1}, n) \rightarrow X_n$ such that the induced map $h_{n*}: H_n(M(G_{n+1}, n)) \rightarrow H_n(X_n)$ is trivial.
- (3) $X = \bigcup_n X_n$.

One sees inductively that X_{n+1} has the desired homology groups by comparing the long exact sequences of the pairs (X_{n+1}, X_n) and (CM, M) where $M = M(G_{n+1}, n)$ and CM is the cone $M \times I / M \times \{0\}$:

$$\begin{array}{ccccccc} 0 & \longrightarrow & H_{n+1}(X_{n+1}) & \longrightarrow & H_{n+1}(X_{n+1}, X_n) & \xrightarrow{\partial} & H_n(X_n) \longrightarrow H_n(X_{n+1}) \longrightarrow 0 \\ & & \uparrow \approx & & \uparrow h_{n*} & & \uparrow \\ & & H_{n+1}(CM, M) & \xrightarrow[\approx]{\partial} & H_n(M) & \approx & G_{n+1} \end{array}$$

The assumption that h_{n*} is trivial means that the boundary map in the upper row is zero, hence $H_{n+1}(X_{n+1}) \approx G_{n+1}$. The other homology groups of X_{n+1} are the same as those of X_n since $H_i(X_{n+1}, X_n) \approx H_i(CM, M)$ for all i by excision, and $H_i(CM, M) \approx \tilde{H}_{i-1}(M)$ since CM is contractible.

In case all the maps h_n are trivial, X is the wedge sum of the Moore spaces $M(G_n, n)$ since in this case the mapping cone construction in (2) produces a wedge sum with the suspension of $M(G_{n+1}, n)$, a Moore space $M(G_{n+1}, n+1)$.

For a space Y , a homotopy equivalence $f: X \rightarrow Y$ where X is constructed as in (1)–(3) is called a **homology decomposition** of Y .

|| **Theorem 4H.3.** *Every simply-connected CW complex has a homology decomposition.*

Proof: Given a simply-connected CW complex Y , let $G_n = H_n(Y)$. Suppose inductively that we have constructed X_n via maps h_i as in (2), together with a map $f: X_n \rightarrow Y$ inducing an isomorphism on H_i for $i \leq n$. The induction can start with X_1 a point since Y is simply-connected. To construct X_{n+1} we first replace Y by the mapping cylinder of $f: X_n \rightarrow Y$, converting f into an inclusion. By the Hurewicz theorem and the homology exact sequence of the pair (Y, X_n) we have $\pi_{n+1}(Y, X_n) \approx H_{n+1}(Y, X_n) \approx H_{n+1}(Y) = G_{n+1}$. We will use this isomorphism to construct a map $h_n: M(G_{n+1}, n) \rightarrow X_n$ and an extension $f: X_{n+1} \rightarrow Y$.

The standard construction of an $M(G_{n+1}, n)$ consists of a wedge of spheres S_α^n corresponding to generators g_α of G_{n+1} , with cells e_β^{n+1} attached according to certain linear combinations $r_\beta = \sum_\alpha n_{\alpha\beta} g_\alpha$ that are zero in G_{n+1} . Under the isomorphism $G_{n+1} \approx \pi_{n+1}(Y, X_n)$ each g_α corresponds to a basepoint-preserving map $f_\alpha: (CS^n, S^n) \rightarrow (Y, X_n)$ where CS^n is the cone on S^n . The restrictions of these f_α 's to S^n define $h_n: \bigvee_\alpha S_\alpha^n \rightarrow X_n$, and the maps $f_\alpha: CS^n \rightarrow Y$ themselves give an extension of $f: X_n \rightarrow Y$ to the mapping cone of $h_n: \bigvee_\alpha S_\alpha^n \rightarrow X_n$. Each relation r_β gives a homotopy $F_\beta: (CS^n, S^n) \times I \rightarrow (Y, X_n)$ from $\sum_\alpha n_{\alpha\beta} f_\alpha$ to the constant map. We use

$F_\beta|S^n \times \{0\}$ to attach e_β^{n+1} , and then $F_\beta|S^n \times I$ gives h_n on e_β^{n+1} and F_β gives an extension of f over the cone on e_β^{n+1} .

This construction assures that $f_*: H_{n+1}(X_{n+1}, X_n) \rightarrow H_{n+1}(Y, X_n)$ is an isomorphism, so from the five-lemma applied to the long exact sequences of these pairs we deduce that $f_*: H_i(X_{n+1}) \rightarrow H_i(Y)$ is an isomorphism for $i \leq n+1$. This finishes the induction step. We may assume the maps f_α and F_β are cellular, so $X = \bigcup_n X_n$ is a CW complex with subcomplexes X_n . Since $f: X \rightarrow Y$ is a homology isomorphism between simply-connected CW complexes, it is a homotopy equivalence. $\square$

As an example, suppose that Y is a simply-connected CW complex having all its homology groups free. Then the Moore spaces used in the construction of X can be taken to be wedges of spheres, and so X_n is obtained from X_{n-1} by attaching an n -cell for each $\mathbb{Z}$ summand of $H_n(Y)$. The attaching maps may be taken to be cellular, making X into a CW complex whose cellular chain complex has trivial boundary maps. Similarly, finite cyclic summands of $H_n(Y)$ can be realized by wedge summands of the form $S^{n-1} \cup e^n$ in $M(H_n(Y), n-1)$, contributing an n -cell and an $(n+1)$ -cell to X . This is Proposition 4C.1, but the present result is stronger because it tells us that a finite cyclic summand of H_n can be realized in one step by attaching the cone on a Moore space $M(\mathbb{Z}_k, n-1)$, rather than in two steps of attaching an n -cell and then an $(n+1)$ -cell.

Exercises

1. Show that if $A \hookrightarrow X$ is a cofibration of compact Hausdorff spaces, then for any space Y , the map $Y^X \rightarrow Y^A$ obtained by restriction of functions is a fibration. [If $A \hookrightarrow X$ is a cofibration, so is $A \times Y \hookrightarrow X \times Y$ for any space Y .]
2. Consider a pushout diagram as at the right, where $B \sqcup_f X$ is B with X attached along A via f . Show that if $A \hookrightarrow X$ is a cofibration, so is $B \hookrightarrow B \sqcup_f X$.

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow & & \downarrow \\ X & \longrightarrow & B \sqcup_f X \end{array}$$
3. For fibrations $E_1 \rightarrow B$ and $E_2 \rightarrow B$, show that a fiber-preserving map $E_1 \rightarrow E_2$ that is a homotopy equivalence is in fact a fiber homotopy equivalence. [This is dual to Proposition 0.19.]
4. Define the dual of an iterated mapping cylinder precisely, in terms of maps from Δ^n , and use this to give a definition of ∇X , the dual of ΔX , for X a complex of spaces.

4.I Stable Splittings of Spaces

It sometimes happens that suspending a space has the effect of simplifying its homotopy type, as the suspension becomes homotopy equivalent to a wedge sum of